

Q1) Given

$$\sum_{i=1}^{100} (3a_i + 17) = 4,175 \quad \text{and} \quad \sum_{j=0}^{98} (a_{j+1} + 21) = 2,909$$

Find a_{100}

$$\sum_{i=1}^{100} (3a_i + 17) = 4,175$$

$$3 \sum_{i=1}^{100} a_i + \sum_{i=1}^{100} 17 = 4,175$$

$$3 \sum_{i=1}^{100} a_i + (100)(17) = 4,175$$

$$3 \sum_{i=1}^{100} a_i + 1,700 = 4,175$$

$$\sum_{i=1}^{100} a_i = \frac{4,175 - 1,700}{3}$$

$$\sum_{i=1}^{100} a_i = 825$$

$$\sum_{j=0}^{98} (a_{j+1} + 21) = 2,909$$

$$\sum_{j=0}^{98} a_{j+1} + \sum_{j=0}^{98} 21 = 2,909$$

$$\sum_{j=0}^{98} a_{j+1} + (99)(21) = 2,909$$

$$\sum_{j=1}^{99} a_j = 2,909 - (99)(21)$$

$$\sum_{j=1}^{99} a_j = 830$$

Combining the two results:

$$a_{100} = \sum_{i=1}^{100} a_i - \sum_{j=1}^{99} a_j$$

$$a_{100} = 825 - 830$$

$$a_{100} = -5$$

Write Final Answer here: $a_{100} = -5$

For **Q2)**, **Q3)** and **Q4)** use the following formulas as necessary:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

*** In order to receive marks your approach must rely on sigma notation, summation properties and usage of given formula(s). *** Answers based on combinatorics will not be accepted.

Q2) Consider the following programming segment:

```
time = 555;
for (i = 1; i <= n; i++){
    time = time + 12;
    for (k = 4; k <= i+6; k++){
        time = time + 24k;
    }
}
```

a) Find the value of the variable `time` (in terms of n) after the programming segment is executed. Your formula must be in the simplest form, fully expanded and terms combined.

(Eg. $time = 7n^2 + 4n + 9$)

$$\begin{aligned}
 time &= 555 + \sum_{i=1}^n \left(12 + \sum_{k=4}^{i+6} 24k \right) = 555 + \sum_{i=1}^n \left(12 + 24 \left(\sum_{k=1}^{i+6} k - \sum_{k=1}^3 k \right) \right) \\
 &= 555 + \sum_{i=1}^n \left(12 + 24 \left(\frac{(i+6)(i+7)}{2} - \frac{(3)(4)}{2} \right) \right) = 555 + \sum_{i=1}^n (12 + 12(i^2 + 13i + 30)) \\
 &= 555 + \sum_{i=1}^n (12i^2 + 156i + 372) = 555 + \sum_{i=1}^n 12i^2 + \sum_{i=1}^n 156i + \sum_{i=1}^n 372 \\
 &= 555 + 12 \sum_{i=1}^n i^2 + 156 \sum_{i=1}^n i + n(372) = 555 + (12) \frac{n(n+1)(2n+1)}{6} + (156) \frac{n(n+1)}{2} + 372n \\
 &= 555 + (2n)(2n^2 + 3n + 1) + (78)(n^2 + n) + 372n \\
 &= 555 + 4n^3 + 6n^2 + 2n + 78n^2 + 78n + 372n \\
 \therefore time &= 4n^3 + 84n^2 + 452n + 555
 \end{aligned}$$

For $n = 50$

$$time = 4(50)^3 + 84(50)^2 + 452(50) + 555 = 733,155$$

$$\sum_{i=1}^n () -$$

Q3) Consider the following programming segment:

```
time = 850;
for (i = 1; i <= n; i++){
    time = time + 10;
    for (k = 9; k <= i+3; k++){
        time = time + 24k;
    }
}
```

- a) Find the value of the variable `time` (in terms of `n`) after the programming segment is executed. Your formula must be in the simplest form, fully expanded and terms combined. (Eg. $time = 7n^2 + 4n + 9$)

Note that the inner loop starts executing for $i = 6$, adjust the bound as follows

$$\begin{aligned}
 time &= 850 + \sum_{i=1}^n 10 + \sum_{i=6}^n \sum_{k=9}^{i+3} 24k = 850 + (n)10 + \sum_{i=6}^n \left(24 \sum_{k=9}^{i+3} k \right) \\
 &= 850 + 10n + \sum_{i=6}^n \left(24 \sum_{k=1}^{i+3} k - 24 \sum_{k=1}^8 k \right) = 850 + 10n + \sum_{i=6}^n \left((24) \frac{(i+3)(i+4)}{2} - (24) \frac{(8)(9)}{2} \right) \\
 &= 850 + 10n + \sum_{i=6}^n (12(i^2 + 7i + 12) - 864) = 850 + 10n + \sum_{i=6}^n (12i^2 + 84i - 720) \\
 &= 850 + 10n + \sum_{i=6}^n 12i^2 + \sum_{i=6}^n 84i - \sum_{i=6}^n 720 = 850 + 10n + 12 \sum_{i=6}^n i^2 + 84 \sum_{i=6}^n i - (n-6+1)720 \\
 &= 850 + 10n + 12 \sum_{i=1}^n i^2 - 12 \sum_{i=1}^5 i^2 + 84 \sum_{i=1}^n i - 84 \sum_{i=1}^5 i - (n-5)720 \\
 &= 850 + 10n + (12) \frac{n(n+1)(2n+1)}{6} - (12) \frac{5(6)(11)}{6} + (84) \frac{n(n+1)}{2} - (84) \frac{(5)(6)}{2} - 720n \\
 &\quad + 3,600 \\
 &= 850 + 10n + (2n)(2n^2 + 3n + 1) - 660 + (42)(n^2 + n) - 1,260 - 720n + 3,600 \\
 &= 850 + 10n + 4n^3 + 6n^2 + 2n - 660 + 42n^2 + 42n - 1,260 - 720n + 3,600 \\
 \therefore time &= 4n^3 + 48n^2 - 666n + 2,530
 \end{aligned}$$

[The formula works for $n \geq 6$. When $1 \leq n \leq 5$, inner loop does not execute so $time = 850 + 10n$]

For $n = 50$

$$time = 4(50)^3 + 48(50)^2 - 666(50) + 2,530 = 589,230$$

$$\sum_{i=1}^n () -$$

Q4) Consider the following programming segment:

```
comp = 10
time = 20;
for (i = 1; i <= n; i++){
    comp = comp + 12;
    for (k = i+1; k <= n; k++){
        time = time + comp + 13;
    }
}
```

- a) Find the value of the variable `time` (in terms of `n`) after the programming segment is executed. Your formula must be in the simplest form, fully expanded and terms combined. (Eg. $time = 7n^2 + 4n + 9$)

Note that at the i -th iteration of the outer loop, the variable `comp` is

$$comp = 10 + \sum_{i=1}^i 12 = 10 + 12i$$

Hence, we can think of the code as

```
comp = 10
time = 20;
for (i = 1; i <= n; i++){
    comp = 10 + 12i;
    for (k = i+1; k <= n; k++){
        time = time + comp + 13;
    }
}
```

Or

```
comp = 10
time = 20;
for (i = 1; i <= n; i++){
    comp = 10 + 12i;
    for (k = i+1; k <= n; k++){
        time = time + 10 + 12i + 13;
    }
}
```

The last iteration of the inner loop is $i=n-1$

$$\begin{aligned} time &= 20 + \sum_{i=1}^{n-1} \left(\sum_{k=i+1}^n (10 + 12i + 13) \right) = 20 + \sum_{i=1}^{n-1} \left(\sum_{k=i+1}^n (12i + 23) \right) \\ &= 20 + \sum_{i=1}^{n-1} (n - (i + 1) + 1)(12i + 23) = 20 + \sum_{i=1}^{n-1} (n - i)(12i + 23) = 20 + \sum_{k=1}^{n-1} (12ni + 23n - 12i^2 - 23i) \\ &= 20 + \sum_{k=1}^{n-1} ((12n - 23)i + 23n - 12i^2) = 20 + \sum_{k=1}^{n-1} (12n - 23)i + \sum_{k=1}^{n-1} (23n) - \sum_{k=1}^{n-1} 12i^2 \end{aligned}$$

$$\sum_{i=1}^n () -$$

$$= 20 + \sum_{k=1}^{n-1} (12n - 23)i + \sum_{k=1}^{n-1} (23n) - \sum_{k=1}^{n-1} 12i^2 = 20 + (12n - 23) \sum_{k=1}^{n-1} i + (n - 1)(23n) - 12 \sum_{k=1}^{n-1} i^2$$

$$= 20 + (12n - 23) \frac{(n-1)n}{2} + (n-1)(23n) - 12 \frac{(n-1)(n)(2n-1)}{6}$$

$$= 20 + (12n - 23) \frac{n^2 - n}{2} + 23n^2 - 23n - (2n - 2)(2n^2 - n)$$

$$= 20 + 6n^3 - 6n^2 - \frac{23}{2}n^2 + \frac{23}{2}n + 23n^2 - 23n - 4n^3 + 6n^2 - 2n$$

$$= 2n^3 + \frac{23}{2}n^2 - \frac{27}{2}n + 20$$

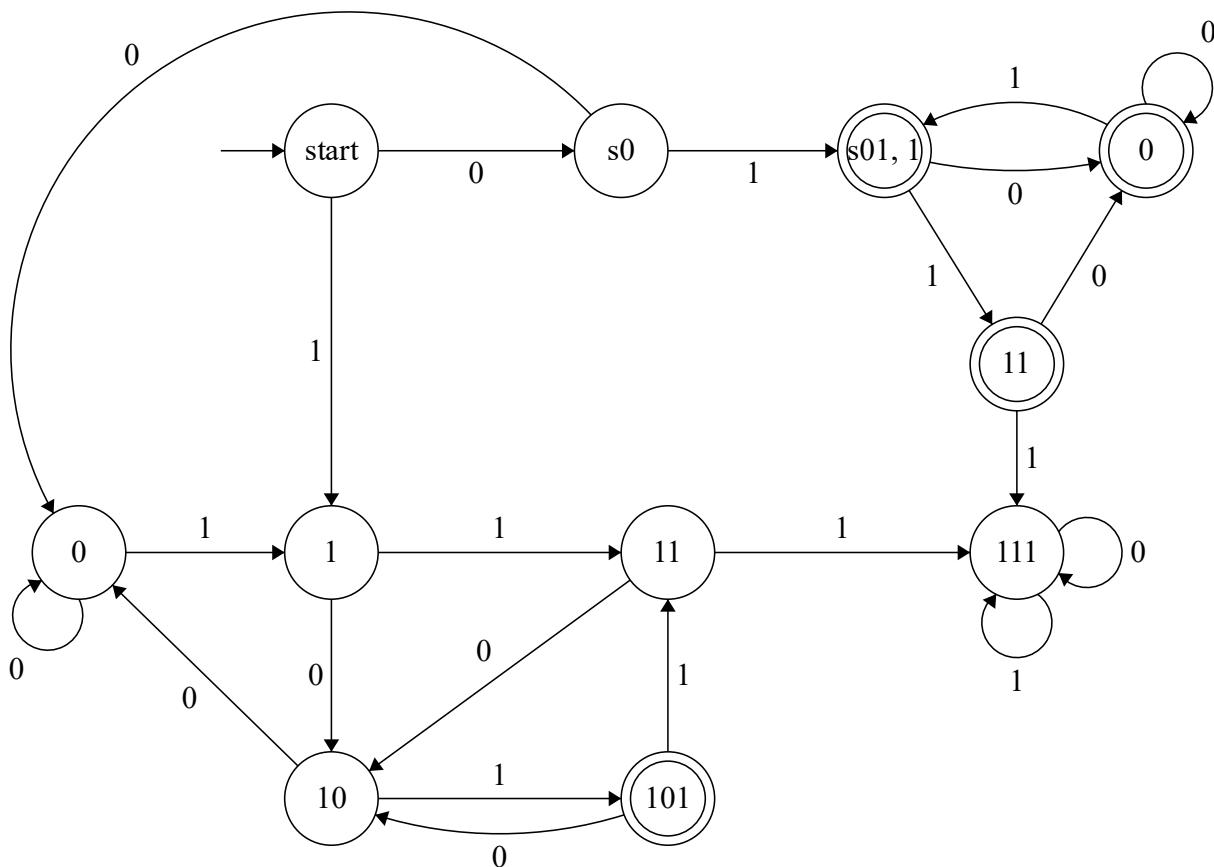
For n = 50

$$time = 2(50)^3 + \frac{23}{2}(50)^2 - \frac{27}{2}(50) + 20 = 278,095$$

$$\sum_{i=1}^n () -$$

- FSM must be clearly labeled and nicely drawn – will not be marked otherwise.
- Make sure there are outgoing 0 and 1 from every state. If that is missing, the machine cannot be marked.
- FSM needs to have the minimum number of required states.
- Properly name the states; S0, S1, S2 or such are not acceptable.
- The start state has to have an arrow pointing to it.
- Effort has to be made that arcs do not intersect or have intersections reduced at the minimum. It is easy to move the states around and check.
- Marks will be taken away for missing details.

Q5) Design a finite state machine with input alphabet {0, 1} that accepts all strings that start with 01 or end with 101, but do not contain 111.



$$\sum_{i=1}^n () -$$

- FSM must be clearly labeled and nicely drawn – will not be marked otherwise.
- Make sure there are outgoing 0 and 1 from every state. If that is missing, the machine cannot be marked.
- FSM needs to have the minimum number of required states.
- Properly name the states; S0, S1, S2 or such are not acceptable.
- The start state has to have an arrow pointing to it.
- Effort has to be made that arcs do not intersect or have intersections reduced at the minimum. It is easy to move the states around and check.
- Marks will be taken away for missing details.

Q6) Design a finite state machine with input alphabet $\{0, 1\}$ that accepts all strings that have an odd number of 0's and the last two bits different from each other.

